① All the organ systems → Respiratory system
→ Digestive system
→ circulatory system.
→ Excretory system.
& int./ext. Organs → liver, brains
perform their own specific functions.

② However, their functions are in complete co-ordination.

1. Respiratory system
F:- Inhale O_2
2. circulatory system
F:- Circulate O_2 & nutrients
3. Excretory system
F:- Remove wastes (Urine)
Digestive system
F:- Provide nutrients

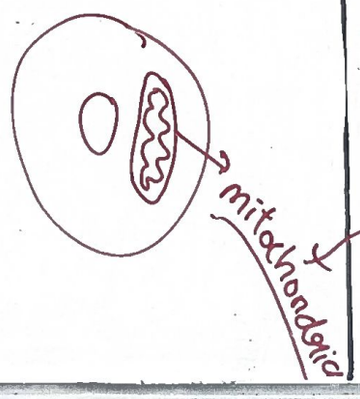
Co-ordination

③ Needs continuos supply of ENERGY to keep these functions running

Step by step process to haved ENERGY :→

1) Main energy sources
carbohydrates, proteins, fats

2) Mitochondria :→ Power house of the cell
→ Energy is havested here

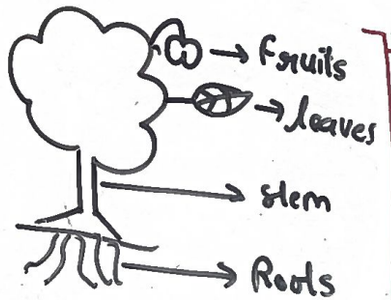


3) Materials necessary for energy production
1. Food stuffs [Carbs, Protein, fats]
2. O_2
circulatory system.
→ Cell

4) This overall process is co-ordinated by
Neural control system
Control system
Endocrine control system.

5) The energy production itself needs energy

How do we get these food stuffs ..... ?



② Remaining food are stored here.

③ Heterotrophs :- Human get food from here (चोर)

① Autotrophs plants :-
→ Synthesize their own food
→ Use some food for itself.

38 ATP → ATP

1 gm Carbs → 4 kCal energy

1 unit of energy = 1 kCal

Chem. Energy (Coal) → Electrical energy

AEROBIC RESPIRATION:→

1 GLYCOLYSIS (O_2) → Glucose → Breakdown

① First step of aerobic respiration.

② Location:- Cytoplasm.

③ Process:- Breakdown/Oxidation of 1 molecule of glucose ($C_6H_{12}O_6$) (6-C)

Product formed:- 1) Pyruvic acid [3-C] :- 2 molecules
2) ATP :- 2 molecules
3) $NADH_2$:- 2 molecules
4) H_2O :- 2 molecules

④ Link reaction:→ 2 Pyruvic acid [3-C] —(link reaction)→ 2 Acetyl Co-A [2-C] → mitochondria

Product formed→ 1) $NADH_2$:- 2 molecules
2) CO_2 :- 2 molecules

⑤ Other name:- EMP pathway.

2 KREBS CYCLE

① Second step

② Location:- Mitochondria

③ Process:-

④ Products:- 1) ATP :- 2 molecules
2) $NADH_2$:- 6 molecules
3) $FADH_2$:- 2 molecules
4) CO_2 :- 4 molecules
5) H_2O :- used up

2 Acetyl CoA + 2 Oxaloacetate → 2 citric acid (carboxylic acid) → $-COOH$

acetyl part is completely oxidized by cyclic chain RKn.

3. ELECTRON TRANSFER CHAIN RKn (ETC RKn)

① 3rd step

Location:- Mitochondria.

Process:- ① 1 $NADH_2$ —(ETC RKn)→ 3 ATP
② 1 $FADH_2$ —(ETC RKn)→ 2 ATP.

Products:- 1) ATPs (main product)
2) H_2O (By product)

Glycolysis → 2 ATP → 2 ATP
4 $NADH_2$ —(×3)→ 12 ATP

Krebs cycle → 2 ATP → 2 ATP
6 $NADH_2$ —(×3)→ 18 ATP
2 $FADH_2$ —(×2)→ 4 ATP

38 ATP

Respiration

Body leve Respiration

Defn
① The exchange of O_2 & CO_2
② blw body & surrounding

Cellular Respiration

Defn
① Step by step oxidation of foodstuffs (Carbs, Proteins, lipids)
② inside the cell
③ to produce energy

Aerobic Respiration

Defn: ①②③ Defn of cellular respiratn
④ with the help of O_2

Steps:-
1. Glycolysis (O_2 ✗)
2. Krebs cycle (O_2 ✓)
3. ETC RKn. (O_2 ✓)

Main source of Energy :- Carbohydrates $[C_nH_{2n}O_2]$ (Glucose, Fructose, Galactose → lactose, sucrose)

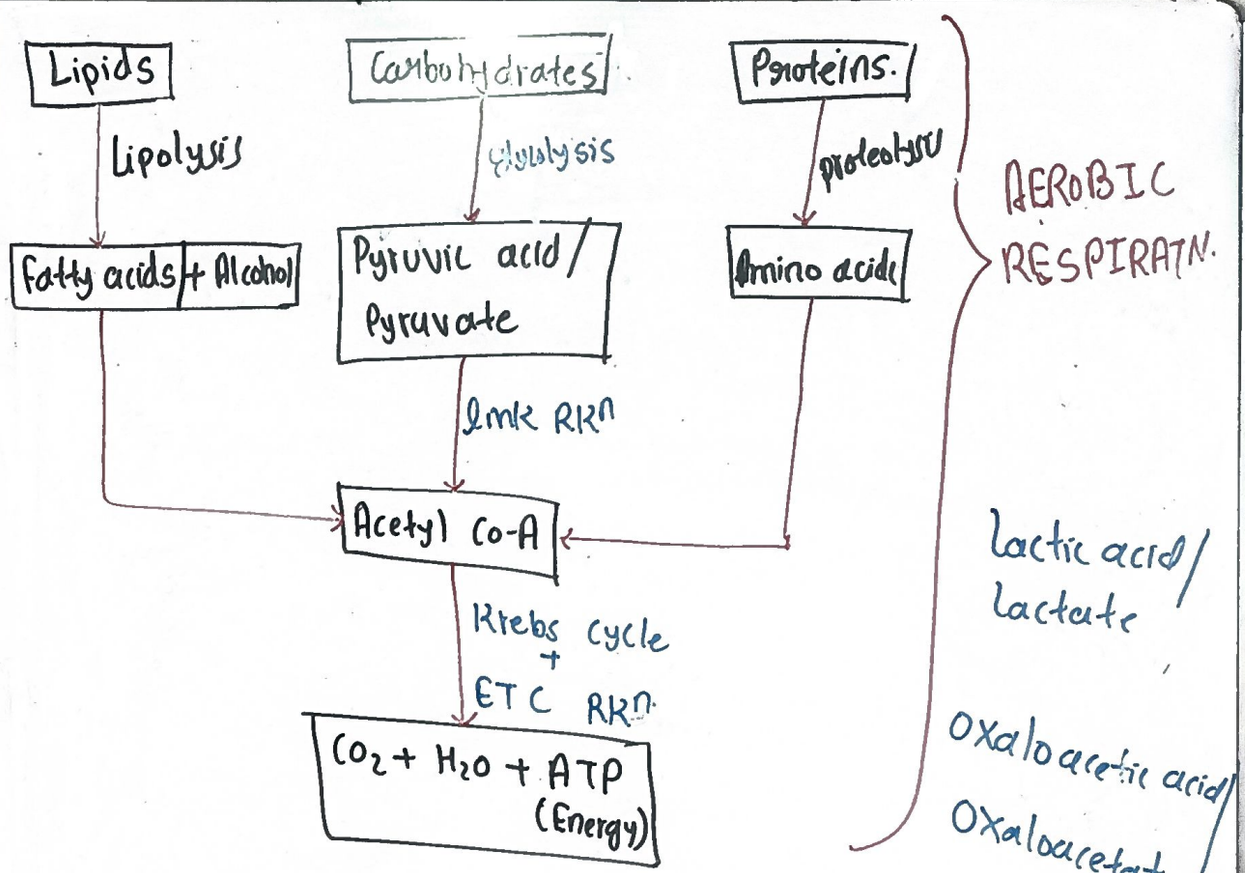
ENERGY PRODUCTION BY PROTEINS & LIPIDS :-

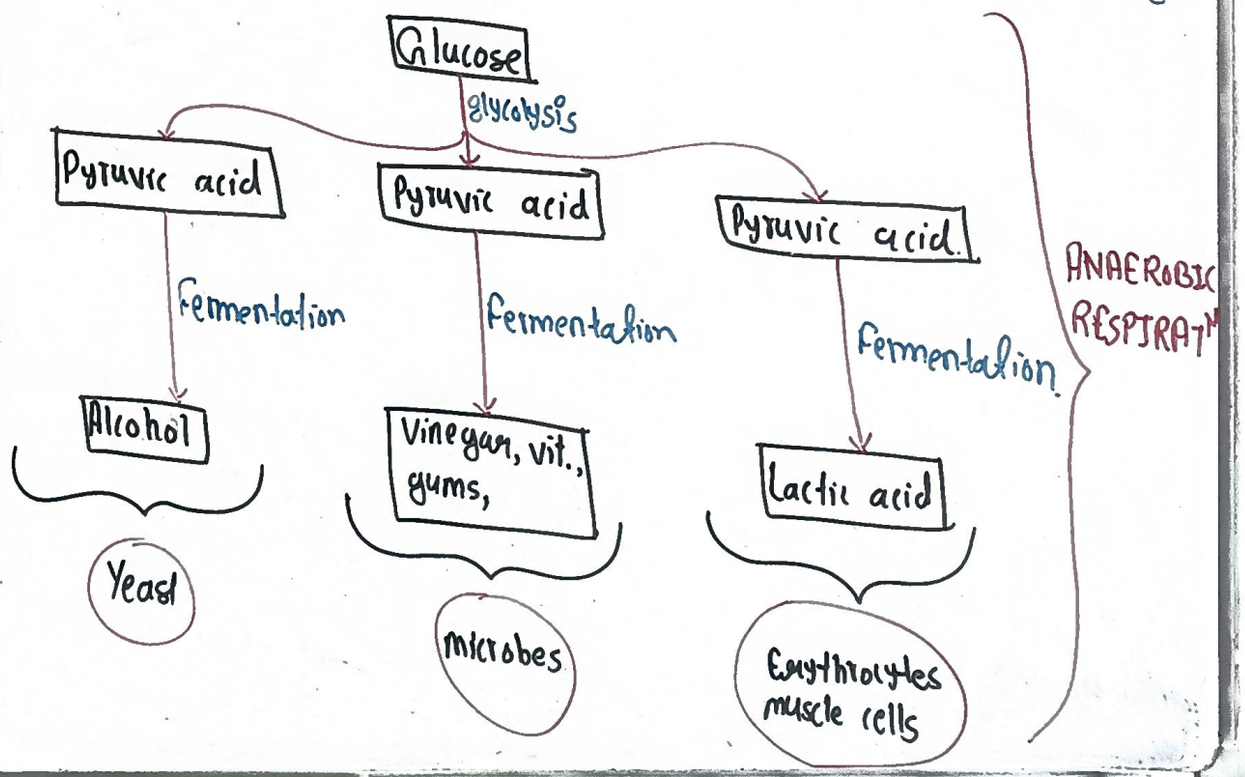
* condition :- 1) Insufficient amount of carbs (Glucose)
2) Hunger OR Fasting

Anaerobic Respiration

Defn: ①②③ Defn of cellular respiratn
④ without the help of O_2

1. Glycolysis
2. Fermentation





ATP (Adenosine Triphosphate) :→

Adenin Ribok

1) Energy currency of the cell

2) **Chemical structure**

① Adenine : A nitrogenous base.

② Ribose :- A pentose (5-carbon) sugar

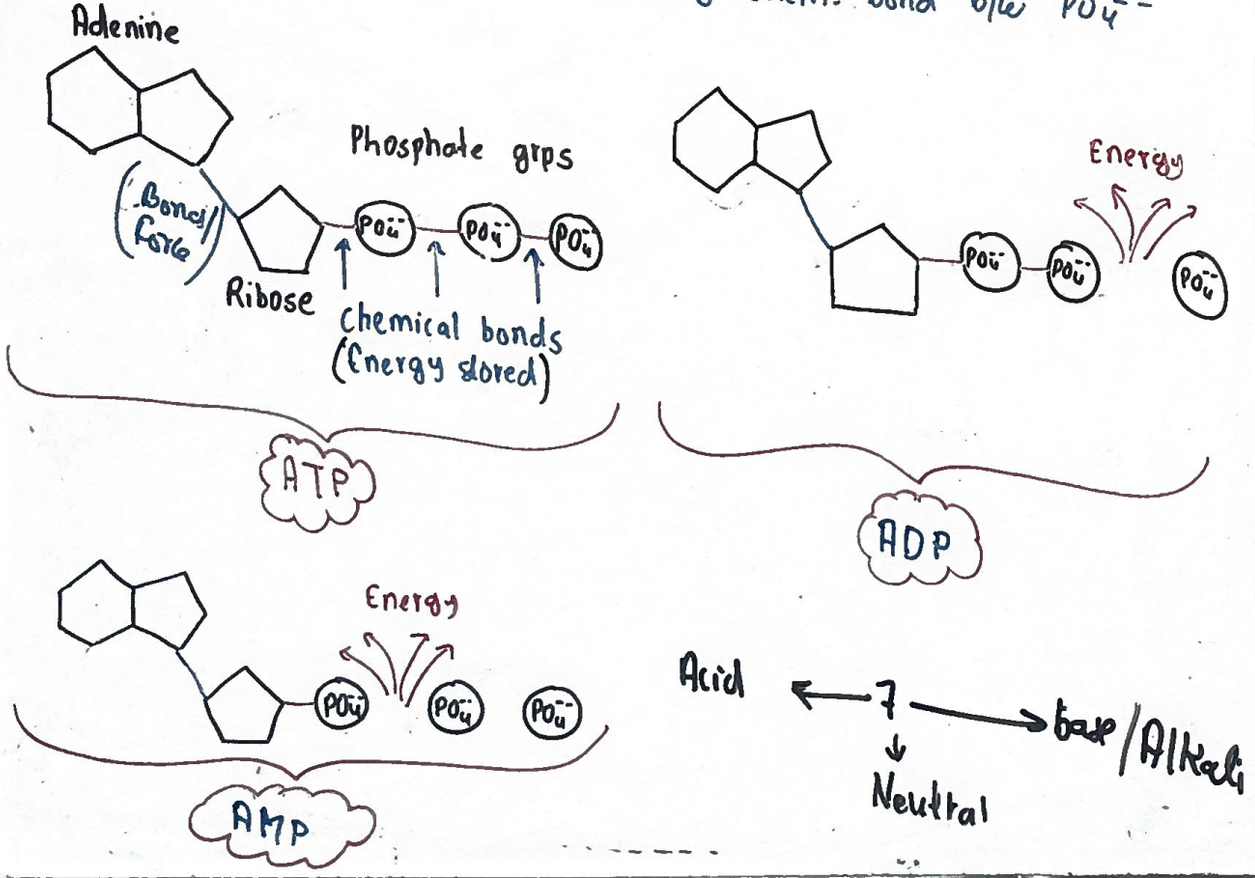
③ Three phosphate (PO_4^{--}) groups

3) **Mechanism**

① Energy is stored within chemical bond b/w PO_4^{--}

② Cells stored these energy rich molecules - ATP.

③ Derived energy by breaking chem. bond b/w PO_4^{--}



Acid ← 7 → base/Alkali

↓ Neutral

ENERGY PRODUCTION IN MICRO-ORGANISMS THROUGH Ana.. Res..

① Conditions :- ① Permanently by organisms that can't survive in O_2
→ eg :- Clostridium, yeasts, Lactobacilli, etc.

Anaerobes

② Temporarily by organisms that survive in O_2 but there is insufficient amount of O_2.
→ eg :→ muscle cells, Erythrocytes } during exercise.
→ Seed submerged in water during germination

② Process :- ① Incomplete oxidation of $C_6H_{12}O_6$.
② ↓ amount of Energy [4 ATP :- Gross / 2 ATP :- Net.]

③ Steps :- 1. Glycolysis :- Glucose → (✓) Pyruvic acid → (✗) Acetyl Co-A

2. Fermentation :- Pyruvic acid → Alcohol (Ethanol), organic acids, Lactic acid, Vinegar (acetate)

Bond formation :- Energy लगती है, Adenosine ribonucleoside
Bond breakdown :- Energy निकलती है, Triphosphate
} Nucleotide

① Nitrogenous base (A, G, C, T.)
② Pentose sugar → Ribose
③ PO_4^{--}
→ Nucleoside → Nucleotide

Adenosine ribonucleoside.

LIPIDS

① Defn:- 1) Substance formed with specific chemical bonds
2) btw fatty acids & alcohol.

② Lipids = fatty acid + alcohol.

③ 9 Kcal / gm lipid (most energy dense nutrient)

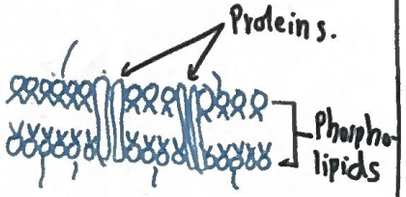
Step by step process:-

1) Digestion of lipids → ① Breakdown
→ ② Absorption (small intestine)

2) Circulated by blood toward cells.

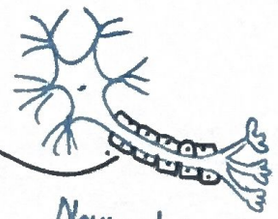
3) Substance made from fatty acids:-

① Phospholipid → plasma membrane.



② Hormones → Progesterone, Estrogen, Testosterone, Aldosterone. } sex hormones

③ Myelin sheath.



Neuron / Nerve cell

↑ lipids → (stored in) → Adipose connective tissue (Intestine)

VITAMINS:-

① Defn:- 1) Group of heterogenous compounds
2) essential for proper operation (functioning)
3) of various life processes
- Blood clotting :- Vit. K
- Vision :- Vit. A
- Hair growth :- Vit. B_{12}
- Immunity :- Vit C.

② No energy obtained from vit.

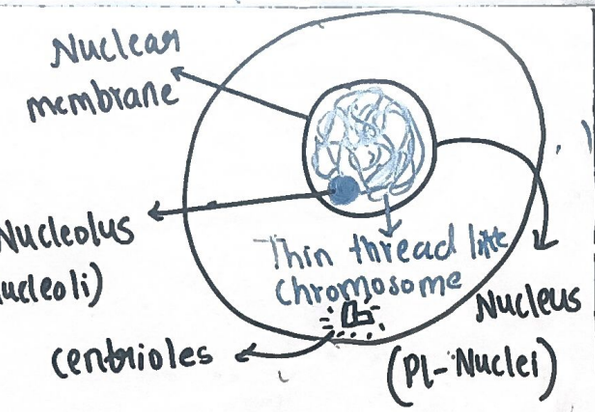
Vitamins
- Water soluble Vitamins
  - Vit. B
  - Vit. C
- Fat (oil) soluble Vitamins
  - Vit. A
  - Vit. D
  - Vit. E
  - Vit. K

* Vit. B_2 (Riboflavin) → $FADH_2$ production
* Vit. B_3 (Nicotinamide) → $NADH_2$ production.

MITOSIS

Karyokinesis

Cytokinesis

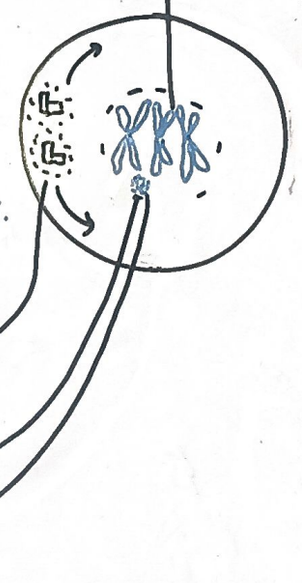


Step 1:- PROPHASE

① Chromosomes :- 1) Thin
2) thread-like } condense.
3) Become short & thick
4) Appear with sister chromatids.

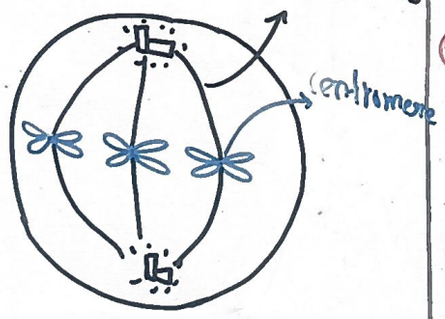
② Centrioles :- 1) Dublicate
2) move to opposite poles of the cell

③ Nuclear membrane, Nucleolus } start to disappear.



Step-2:- METAPHASE

① Nuclear membrane, Nucleolus } completely disappears.



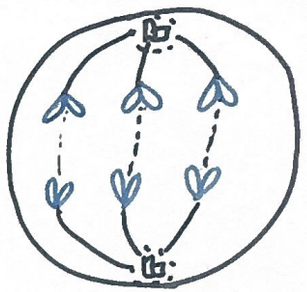
② Chromosome :→ Allign parallel to the cell's equator

Centrioles :- 1) → Spindle fibres form
→ b/w centromes & each centrioles.

Step 3:- ANAPHASE

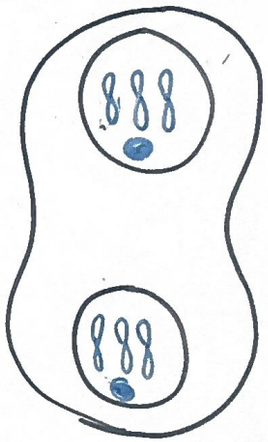
① Centromeres :- Split

② Sister chromatids :- 1) separate
2) Pulled a part
- by spindle fibres
- to opposite poles of the cell
3) Appear like bunch of bananas.



Step-4:- TELOPHASE

① Chromosomes :- 1) Reached opposite poles
2) Two groups of daughter chromo. forms.
3) Again become thin thread like



② Nuclear membrane :- Reappear around each group of daughter chromosome

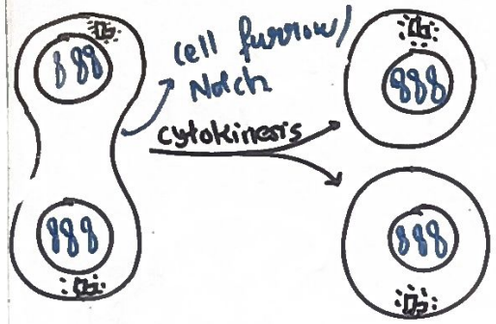
③ Nucleolus :- Reappears in each group of daughter chromosomes

④ Thus, two nuclei forms within the cell.

Cytokinesis :-

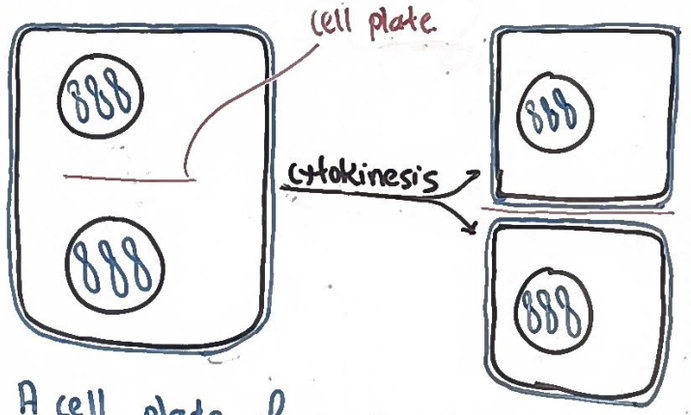
1) The cytoplasm divides to form two new daughter cells.

In animal cell



A notch/cell furrow forms & deepens.
Ultimately it divides entire cytoplasm.

In plant cell



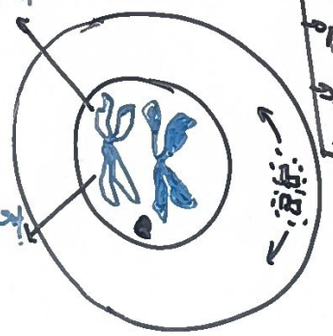
A cell plate form in plant cell it elongates & Ultimately divides entire cytoplasm

MEIOSIS

Step-1 :- LEPTOTENE

① Chromosomes begins to condense

② Chromosomes become thick & short

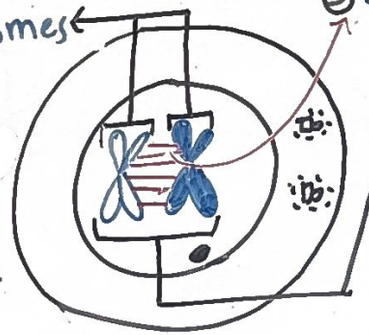


Meiosis
- Meiosis-I
  - Prophase-I
  - Metaphase-I
  - Anaphase-I
  - Telophase-I
- Meiosis-II
  - Prophase-II
  - Metaphase-II
  - Anaphase-II
  - Telophase-II

Step-2] ZYGOTENE :-

① Homologous chromosomes pairs together

② This process is called SYNAPSIS.



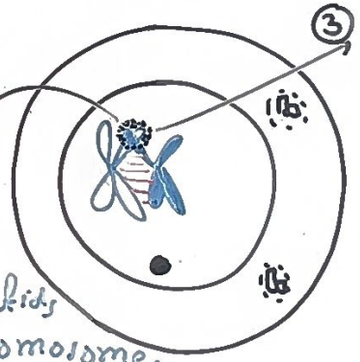
⑤ Synaptonemal complex forms

④ Paired chromosomes
- Bivalent :- 2 chromosomes OR
- Tetrads :- 4 chromatids. → 4

Step-3] PACHYTENE.

① longest step of prophase-I

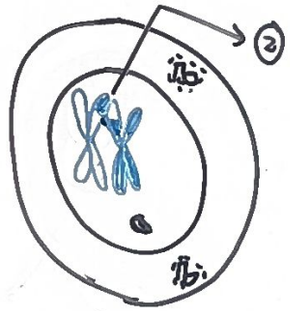
② Crossing over occurs blw non-sister chromatids of homologous chromosomes.



③ Recombination nodule appear at crossing over site

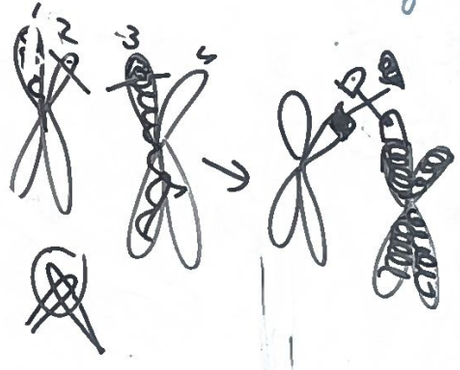
Step-4] DIPLOTENE

① Synaptonemal complex dissolves



② Chiasmata appear at crossing over

X-shaped



Step-5] DIAKINESIS

① Chiasmata terminalise.

② Nuclear membrane & nucleolus disappear

③ Spindle fibres start to form

